11 May 1973
SUPERSEDING
MIL-F-35093A
11 October 1966

MILITARY SPECIFICATION

FOOD COLORING, LIQUID

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers liquid food coloring for use by the Armed Forces as an item of general issue.

1.2 Classification. The product shall be of the following types, as specified (see 6.1):

Type	I	-	Bright red (strawberry shade)
Type	II	-	Egg shade
Type	III	-	Blue
Type	IV	-	Green
Type	V	-	Caramel

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-G-460 - Glass Containers, One Gallon Capacity and Smaller For Other Than Medicinal Products; Packaging and Packing Of.

MILITARY

MIL-L-35078 - Loads, Unit: Preparation of Nonperishable Subsistence.

FSC 8950

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-668 - Sanitary Standards for Food Plants.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

PUBLICATIONS

* U. S. Department of Agriculture

United States Standards for Condition of Food Containers

(Application for copies should be addressed to the Information Division, Agricultural Marketing Service, U. S. Department of Agriculture, Washington, D.C. 20250.)

U. S. Department of Health, Education, and Welfare

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder

(Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, D.C. 20402.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

National Research Council

Food Chemical Codex (F.C.C.)

(Application for copies should be addressed to the Office of Director, National Academy of Science, National Research Council, Washington, D.C. 20418.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

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3. REQUIREMENTS

3.1 Materials. All materials used in the manufacture of liquid food colors shall be edible, clean, sound and of good quality.

3.1.1 Colors, dry. Dry colors, except caramel color, used in making the liquid coloring shall be certified by the Federal Food and Drug Administration. Caramel coloring shall be a dry, readily soluble, brown powder obtained from sucrose or glucose.

* 3.1.2 Preservatives. Preservatives shall be glycerine or propylene glycol with added sodium benzoate. The preservatives shall comply with additive requirements of Food Chemical Codex.

* 3.1.3 Water. The water used shall be distilled or deionized water.

* 3.1.4 Acid. The acidulants shall be citric, malic or tartaric acid and shall conform to the Food Chemical Codex.

* 3.2 Formulations. The weight of colors shall be on an "as certified basis" and formulations shall be as follows:

Type I - Bright red (strawberry)

2.0	-	ounces FD&C Red No. 40
2.9	-	ounces FD&C Red No. 3
0.1	-	ounces FD&C Violet No. 1
32.0 - 48.0	-	ounces glycerine or propylene glycol.
Water to	-	1 gallon

Type II - Egg shade

.125	-	ounces sodium benzoate
.125 - .25	-	ounces citric acid, tartaric acid or malic acid
5.0	-	ounces FD&C Yellow No. 5 (Tartrazine)
1.0	-	ounces FD&C Yellow No. 6 (Sunset yellow)
22.0	-	ounces glycerine or propylene glycol
Water to	-	1 gallon

Type III - Blue

.125	-	ounces sodium benzoate
.125 - .25	-	ounces citric acid, tartaric acid or malic acid
1.75	-	ounces FD&C Blue No. 1 (Brilliant Blue)
22.0	-	ounces glycerine or propylene glycol
Water to	-	1 gallon

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Type IV - Green

.125	-	ounces sodium benzoate
.125 - .25	-	ounces citric acid, tartaric acid or malic acid
1.7	-	ounces FD&C Blue No. 1
3.0	-	ounces FD&C Yellow No. 5
22.0	-	ounces glycerine or propylene glycol
Water to	-	1 gallon

* 3.3 Process. Distilled or deionized water shall be brought to a boil and allowed to cool. Color shall be added slowly and mixed thoroughly. The remainder of the ingredients shall be added, mixed thoroughly; the resultant product shall be clear and devoid of lumps. Any alternative method of mixing ingredients giving equivalent results in the finished product is permissible. The product shall then be bottled.

3.4 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

* 3.5 Workmanship. The product shall be prepared, processed and packaged under modern sanitary conditions and in accordance with good commercial practice. The product shall be processed in establishments meeting the requirements of MIL-STD-668.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 Preacceptance inspection (plant sanitation inspection). The product shall be rejected if produced in plants not meeting the requirements of 3.5.

4.3 Inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.3.1 Component and material inspection. In accordance with 4.1, components and material shall be inspected in accordance with all the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

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* 4.3.1.1 Ingredients. Conformance of ingredients to identity requirements (kind, quality standard) shall be ascertained by examination of pertinent labels, invoices, or other valid documents. In the event water is processed by the supplier, examination shall be performed to determine if the water is being processed by methods which will yield it distilled or demineralized. Nonconformance shall indicate an unacceptable ingredient and use of such shall be cause for rejection of the involved finished product.

4.3.1.2 Examination of dry colors and preservatives. Examination of components for compliance with requirements of 3.1.1 and 3.1.2 shall be accomplished by a review of labels, invoices or other valid documents.

* 4.3.1.3 Acids. Conformance of acids as to kind and quality standards (Food Chemical Codex) as specified in 3.1.4 shall be ascertained by examination of pertinent labels, invoices or other valid documents.

* 4.3.2 Examination of formulation. Examination shall be performed to determine conformance to formulation requirements of 3.2. Formulation records shall be maintained. The records shall be examined by the assigned Government inspector. In the event the Government inspector is in the plant during production, he shall perform an examination of the actual formulation. When specified (see 6.1), the Government inspector shall perform continuous examination of the actual formulation.

* 4.3.3 Examination of end item. Examinations shall be in accordance with tables II and III, 4.3.3.1 thru 4.3.3.3. Sampling plans for the aforementioned tables shall be in accordance with table I. The acceptable quality levels (AQLs) shall be expressed as percent defective.

TABLE I. Sampling plans

Table	Sample unit	Lot size expressed in	Inspection level	AQLs Minor
II	1 filled bottle	Bottles	S-3	2.5
III	1 labelled bottle	Bottles	S-2	2.5

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* TABLE II. Examination for volume 1/

Category	Defects
Minor	
201	Less than specified net volume.

1/ Perform examination when product is at room temperature (65°-75°F), use a suitably graduated cylinder(s) i.e., 1/4 ounce = 7.4 ml; 8 ounce = 236 ml; 16 ounce = 473 ml.

* TABLE III. Examination for label adhesion (applicable to overseas shipment only) 1/

Category	Defects
Minor	
201	Displacement of label.

1/ See 4.4.1 for procedure.

* 4.3.3.1 Examination of primary container. Examination for condition and labeling shall be in accordance with the United States Standards for Condition of Food Containers. Plastic bottles shall be examined for the applicable defects contained in the examination of glass containers. When 1/4 ounce plastic bottles are procured, include the following minor defect: Color of cap does not correspond to contents.

* 4.3.3.2 Examination for packing, marking, shipping containers. Examination shall be in accordance with the quality assurance provisions for the end item in PPP-G-460, except as follows: when examining intermediate package containing 1/4 oz. bottles, include the following major defect: variety of colors not as specified (examine 1 intermediate package per sample unit). In addition, the examination of paperboard boxes for closure, as referenced in PPP-G-460, shall not apply.

* 4.3.3.3 Examination of unit loads. Unit loads shall be examined in accordance with the quality assurance provisions of MIL-L-35078.

* 4.4 Test methods. Examination and test procedures which differ from those specified herein, unless otherwise excepted, may be used by the supplier if they provide a quality assurance equivalent to that specified. If the

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Government contracting officer determines that such procedures and controls do not provide, as a minimum, such quality asssurance, the supplier will use the procedures set forth herein. In case of dispute as to test results, the test methods specified herein will govern.

4.4.1 Water-resistant test for label adhesive. The label adhesive shall be tested for water-resistance as follows: Submerge glass container with label affixed in room temperature water for 4 hours. Remove from water and remove excess water by blotting with paper towel. Press finger firmly against label (in an area where adhesive has been applied) and move finger (firmly pressed against the label) towards the bottom of the bottle.- Displacement of label indicates failure to meet the requirement.

5. PREPARATION FOR DELIVERY

5.1 Packaging, level C. Not less than one-quarter, 8 or 16 fluid ounces of the product, as specified (see 6.1); shall be packaged in commercial glass containers. Alternatively, 1/4 ounce shall be packaged in a plastic bottle. The individual 1/4 ounce containers shall each have a cap in a color corresponding to the color of the contents.

* 5.2 Packing. The product shall be packed in accordance with 5.2.1 or 5.2.2, as applicable.

* 5.2.1 Glass bottles, levels A, B and C. The product in glass bottles shall be packed in accordance with the level A, B or C requirements of PPP-G-460, as specified (see 6.1), except that for 1/4 ounce bottles, 1 bottle each of red, blue, green and egg shade shall be inclosed in any suitable commercial interior package, and 144 interior packages shall be packed in the shipping container, in a suitable arrangement.

* 5.2.2 Plastic bottles, levels A, B and C. The 1/4 ounce plastic bottles, shall be packaged and packed in the assortment, interior package, number and arrangement, specified for glass 1/4 ounce bottles for the level specified (see 6.1).

* 5.3 Unit loads. When specified (see 6.1), the product, packed as specified in 5.2, shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified.

5.4 Labeling and marking.

* 5.4.1 Glass containers for domestic shipment (8 and 16 ounce). Labeling shall be in accordance with the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

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* 5.4.2 Glass containers for overseas shipment (8 and 16 ounce). Labeling information shall be in accordance with 5.4.1. The labeling information shall be printed on a hard sized or varnished label. The label shall be affixed in position with a water-resistant adhesive.

* 5.4.3 Interior package, 1/4 ounce container. The interior package shall be labeled in accordance with the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

5.4.4 Shipping containers. Shipping containers shall be marked in accordance with MIL-STD-129, and with the precautionary marking specified in PPP-G-460.

* 5.4.5 Unit loads. Unit loads shall be marked in accordance with MIL-L-35078.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type of product required (see 1.2).
(c) Size of packaging required (see 5.1).
(d) Level of packing required (see 5.2.1 and 5.2.2).
(e) Type and class of unit load when specified (see 5.3).

* 6.2 Based on the conditions known or expected to be encountered during shipment, handling and storage of the specific item being procured, the contracting officer should select the appropriate level of pack in accordance with the criteria established in AR 700-15, NAVSUPINST 4030.28, AFR 71-6, MCO 4030.14D or DSAR 4145.7, as applicable.

* 6.3 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

Army - GL
Navy - SA
Air Force - 45

Review activities:

Army - MD
Navy - MC, MS
DP - SS

Preparing activity:

Army - GL

Project No. 8950-0212

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

MIL-F-35093B FOOD COLORING, LIQUID

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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